

| SEQ NO | REV | NOTE NO. | ITEM NAME | ITEM CODE  | SIZE RANGE / SMALL END SIZES |      | DESCRIPTION                                                                                     | REFERENCE / LARGE END SIZES |    | NOTES |                                                                                                                                                                            |
|--------|-----|----------|-----------|------------|------------------------------|------|-------------------------------------------------------------------------------------------------|-----------------------------|----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10     | U04 |          |           |            |                              |      | ***** VALVES *****                                                                              |                             |    | B1    | ALL BOLTS ARE METRIC DIA.                                                                                                                                                  |
| 20     | U04 |          |           |            |                              |      | ***** BALL VALVES *****                                                                         |                             |    | FS1   | USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.                                                                                            |
| 30     |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 30     | U04 |          | BALL      | A01AC01    | 1                            | 4    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO                                  | 0                           | 0  |       |                                                                                                                                                                            |
| 40     | U04 |          | BALL      | A01AC03    | 6                            | 6    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB GO                                  | 0                           | 0  | FV5   | WHERE THERE IS A SIZE CHANGE FROM 18" - 24" CARBON STEEL TO VICTAULIC SYSTEM, USE REDUCER AND STANDARD RF WNK FLANGE TO CONNECT TO FLANGE ADAPTER NIPPLE ON VICTAULIC END. |
| 50     | U04 |          | BALL      | A01AC07    | 8                            | 8    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO                                  | 0                           | 0  |       |                                                                                                                                                                            |
| 60     | U04 | VU2      | BALL      | BA6B2      | 0.5                          | 0.5  | BALL 150# SCRD FNPT BS1400 LG2 BODY AND TRIM FB LO                                              | 0                           | 0  |       |                                                                                                                                                                            |
| 70     | U04 |          | BALL      | BA6B2      | 0.75                         | 2    | BALL 150# SCRD FNPT BS1400 LG2 BODY AND TRIM FB LO                                              | 0                           | 0  |       |                                                                                                                                                                            |
| 80     | U04 | VA5      | BALL      | B01AC01    | 1                            | 3    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO                                  | 0                           | 0  |       |                                                                                                                                                                            |
| 90     | U04 | VA5      | BALL      | B01AC03    | 4                            | 4    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO                                  | 0                           | 0  | FV6   | COMPONENTS AT NPD 12", 14" AND 16" ARE TO MEET MANUFACTURER'S MINIMUM WALL THICKNESS REQUIREMENTS OF STANDARD WT.                                                          |
| 100    | U04 | VA5      | BALL      | B01AC07    | 6                            | 12   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO                                  | 0                           | 0  |       |                                                                                                                                                                            |
| 110    | U04 | VA5      | BALL      | B03AC01    | 0.75                         | 1    | BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO                                  | 0                           | 0  |       |                                                                                                                                                                            |
| 120    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 120    | U04 |          |           |            |                              |      | ***** BUTTERFLY VALVES *****                                                                    |                             |    | NR1   | SIZES SHOWN IN SIZE COLUMNS ARE THE SMALL END OF REDUCERS AND THE BRANCH SIZE OF OLETS AND TEES.                                                                           |
| 130    | U04 |          | BUTTERFLY | T01AC61    | 10                           | 24   | BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO                       | 0                           | 0  | PA6   | PIPE IS SUPPLIED WITH STANDARD PLAIN END/ BW END AND GROOVE TO BE CUT DURING FABRICATION.                                                                                  |
| 140    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 140    | U04 |          |           |            |                              |      | ***** GATE VALVES *****                                                                         |                             |    | UD7   | USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST                                                                     |
| 150    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 150    | U04 |          | GATE      | GA6B1      | 0.75                         | 1    | GATE 150# SCRD FNPT BS1400 LG2 BODY AND TRIM HW                                                 | 0                           | 0  |       |                                                                                                                                                                            |
| 160    | U04 |          | GATE      | G01AC48    | 1.5                          | 1.5  | GATE 150# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW                                                | 0                           | 0  |       |                                                                                                                                                                            |
| 170    | U04 |          | GATE      | G01AC41    | 2                            | 8    | GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW                                                     | 0                           | 0  | VA5   | USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS                                                                                                    |
| 180    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 180    | U04 |          |           |            |                              |      | ***** GLOBE VALVES *****                                                                        |                             |    | VC1   | SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.                                    |
| 190    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 190    | U04 |          | GLOBE     | LA6B1      | 0.75                         | 1    | GLOBE 150# SCRD FNPT BS1400 LG2 BODY AND TRIM HW                                                | 0                           | 0  |       |                                                                                                                                                                            |
| 200    | U04 |          | GLOBE     | L01AC48    | 1.5                          | 1.5  | GLOBE 150# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW                                                | 0                           | 0  |       |                                                                                                                                                                            |
| 210    | U04 |          | GLOBE     | L01AC41    | 2                            | 6    | GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW                                                     | 0                           | 0  | VU2   | USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY.                                                                                                                           |
| 220    | U04 |          | GLOBE     | L01AC42    | 8                            | 10   | GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO                                                     | 0                           | 0  |       |                                                                                                                                                                            |
| 230    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 230    | U04 |          |           |            |                              |      | ***** CHECK VALVES *****                                                                        |                             |    |       |                                                                                                                                                                            |
| 240    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 240    | U04 | VC1      | CHECK     | CA6B1      | 0.75                         | 1    | CHECK 150# SCRD FNPT BS1400 LG2 BODY AND TRIM SPRING PISTON                                     | 0                           | 0  |       |                                                                                                                                                                            |
| 250    | U04 | VC1      | CHECK     | C01AC50    | 1.5                          | 1.5  | CHECK PISTON 150# RF LTCS A350 LF2 CL.1 TRIM 16                                                 | 0                           | 0  |       |                                                                                                                                                                            |
| 260    | U04 |          | CHECK     | C01AC45    | 2                            | 6    | CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED                                 | 0                           | 0  |       |                                                                                                                                                                            |
| 270    | U04 |          | CHECK     | C01AC52    | 8                            | 16   | CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16                                                | 0                           | 0  |       |                                                                                                                                                                            |
| 280    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 280    | U04 |          |           |            |                              |      | ***** PIPE *****                                                                                |                             |    |       |                                                                                                                                                                            |
| 290    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 290    | U04 |          | PIPE      | 5358601    | 0.75                         | 1    | PIPE SCH 160 SMLS LTCS A333 GR.6 GALV A153 T&C                                                  | 0                           | 0  |       |                                                                                                                                                                            |
| 300    | U04 | PA6      | PIPE      | 5358507_VC | 1.5                          | 3    | PIPE SCH 80 SMLS LTCS A333 GR. 6 PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)  | 0                           | 0  |       |                                                                                                                                                                            |
| 310    | U04 | PA6      | PIPE      | 5358505_VC | 4                            | 10   | PIPE SCH STD SMLS LTCS A333 GR. 6 PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION) | 0                           | 0  |       |                                                                                                                                                                            |
| 320    | U04 | PA6      | PIPE      | 5358502_VC | 12                           | 12   | PIPE SCH 30 SMLS LTCS A333 GR. 6 PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)  | 0                           | 0  |       |                                                                                                                                                                            |
| 330    | U04 | PA6      | PIPE      | 5358501_VC | 14                           | 16   | PIPE SCH 20 SMLS LTCS A333 GR. 6 PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)  | 0                           | 0  |       |                                                                                                                                                                            |
| 340    | U04 |          | PIPE      | 5358597    | 18                           | 24   | PIPE SCH 20 EFW LTCS A671 C060 CL.22 BE                                                         | 0                           | 0  |       |                                                                                                                                                                            |
| 350    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 350    | U04 | NR1, FV5 |           |            |                              |      | ***** FITTINGS *****                                                                            |                             |    |       |                                                                                                                                                                            |
| 360    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 360    | U04 |          | ELBOW     | 5491290    | 0.75                         | 1    | ELL 90 DEG 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153                                         | 0                           | 0  |       |                                                                                                                                                                            |
| 370    | U04 |          | ELBOW     | 5491300    | 0.75                         | 1    | ELL 45 DEG 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153                                         | 0                           | 0  |       |                                                                                                                                                                            |
| 380    | U04 |          | ELBOW     | 5392280    | 18                           | 24   | ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W                                                        | 0                           | 0  |       |                                                                                                                                                                            |
| 390    | U04 |          | ELBOW     | 5392281    | 18                           | 24   | ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W                                                        | 0                           | 0  |       |                                                                                                                                                                            |
| 400    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 400    | U04 |          | TEE       | 5491310    | 0.75                         | 1    | TEE 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153                                                | 0                           | 0  |       |                                                                                                                                                                            |
| 410    | U04 |          | TEE       | 5392295    | 18                           | 24   | TEE BW SCH 20 LTCS A420 WPL6-W                                                                  | 0                           | 0  |       |                                                                                                                                                                            |
| 420    | U04 |          | TEE RED   | 5491311    | 0.75                         | 0.75 | TEE RED 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153                                            | 1                           | 1  |       |                                                                                                                                                                            |
| 430    | U04 |          | TEE RED   | 5392296    | 8                            | 10   | TEE RED BW SCH 20/STD LTCS A420 WPL6-W                                                          | 18                          | 24 |       |                                                                                                                                                                            |
| 440    | U04 |          | TEE RED   | 5392296    | 14                           | 16   | TEE RED BW SCH 20/STD LTCS A420 WPL6-W                                                          | 18                          | 24 |       |                                                                                                                                                                            |
| 450    | U04 |          | TEE RED   | 5392244    | 12                           | 12   | TEE RED BW SCH 20/SCH 30 LTCS A420 WPL6-W                                                       | 18                          | 24 |       |                                                                                                                                                                            |
| 460    | U04 |          | TEE RED   | 5392292    | 18                           | 20   | TEE RED BW SCH 20 LTCS A420 WPL6-W                                                              | 20                          | 24 |       |                                                                                                                                                                            |
| 470    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 470    | U04 |          | CAP       | 5491330    | 0.75                         | 1    | CAP 3000# SCRD FNPT LTCS A350 LF2 CL.1 GALV A153                                                | 0                           | 0  |       |                                                                                                                                                                            |
| 480    | U04 |          | CAP       | 5392286    | 18                           | 24   | CAP BW SCH 20 LTCS A420 WPL6                                                                    | 0                           | 0  |       |                                                                                                                                                                            |
| 490    |     |          |           |            |                              |      |                                                                                                 |                             |    |       |                                                                                                                                                                            |
| 490    | U04 |          | RED CONC  | 5392297    | 10                           | 10   | REDUCER CONC SCH 20/STD LTCS A420 WPL6-W                                                        | 18                          | 18 |       |                                                                                                                                                                            |
| 500    | U04 |          | RED CONC  | 5392243    | 12                           | 12   | REDUCER CONC SCH 20/SCH 30 LTCS A420 WPL6-W                                                     | 18                          | 20 |       |                                                                                                                                                                            |
| 510    | U04 |          | RED CONC  | 5392287    | 14                           | 20   | REDUCER CONC SCH 20 LTCS A420 WPL6-W                                                            | 18                          | 24 |       |                                                                                                                                                                            |
| 520    | U04 |          | RED ECC   | 5392298    | 10                           | 10   | REDUCER ECC SCH 20/STD LTCS A420 WPL6-W                                                         | 18                          | 18 |       |                                                                                                                                                                            |

|                                                                                    |                                               |                                                                                                                                                                       |  |                  |  |                                                                         |  |                                                                                                |  |            |            |     |                |                   |
|------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------|--|-------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------|--|------------|------------|-----|----------------|-------------------|
|  | TENGIZCHEVROI<br>TCO<br>TENGIZ,<br>KAZAKHSTAN | CONSTRUCTION                                                                                                                                                          |  | CORR.ALLOW.      |  | SERVICE                                                                 |  | MATERIAL                                                                                       |  | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|                                                                                    |                                               | 3/4"- 1" - SCREWED<br>1 1/2" - 16" - VICTAULIC GROOVE & COUPLING<br>18" & ABOVE BUTTWELDED<br>SEAL WELDING - NONE<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE |  | CORR. ALLOW. 3MM |  | DESIGN: 15.8 BARG<br>CODE - ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO |  | MATERIAL: LTCS VICTAULIC COUPLINGS<br>FLANGE RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 110 °C |  | 12/16/2024 | BY<br>OZAC | U04 | Page 1<br>of 3 | 150VC01           |
|                                                                                    |                                               |                                                                                                                                                                       |  |                  |  |                                                                         |  |                                                                                                |  |            |            |     |                |                   |

| SEQ NO | REV | NOTE NO. | ITEM NAME     | ITEM CODE | SIZE RANGE / SMALL END SIZES |      | DESCRIPTION                                                           | REFERENCE / LARGE END SIZES |     | NOTES |
|--------|-----|----------|---------------|-----------|------------------------------|------|-----------------------------------------------------------------------|-----------------------------|-----|-------|
| 530    | U04 |          | RED ECC       | 5392242   | 12                           | 12   | REDUCER ECC SCH 20/SCH 30 LTCS A420 WPL6-W                            | 18                          | 20  |       |
| 540    | U04 |          | RED ECC       | 5392288   | 14                           | 20   | REDUCER ECC SCH 20 LTCS A420 WPL6-W                                   | 18                          | 24  |       |
| 550    |     |          |               |           |                              |      | ***** FITTINGS - VICTAULIC *****                                      |                             |     |       |
| 550    | U04 |          |               |           |                              |      |                                                                       |                             |     |       |
| 560    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 560    | U04 |          | ELBOW         | 5484860   | 1.5                          | 3    | ELL 90 DEG GRV SCH 80 LTCS A352 GR LCC GALV                           | 0                           | 0   |       |
| 570    | U04 |          | ELBOW         | 5484861   | 4                            | 10   | ELL 90 DEG GRV SCH STD LTCS A352 GR LCC GALV                          | 0                           | 0   |       |
| 580    | U04 |          | ELBOW         | 5484865   | 1.5                          | 3    | ELL 45 DEG GRV SCH 80 LTCS A352 GR LCC GALV                           | 0                           | 0   |       |
| 590    | U04 |          | ELBOW         | 5484866   | 4                            | 10   | ELL 45 DEG GRV SCH STD LTCS A352 GR LCC GALV                          | 0                           | 0   |       |
| 600    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 600    | U04 |          | TEE           | 5485120   | 1.5                          | 3    | TEE GRV SCH 80 LTCS A352 GR LCC GALV                                  | 0                           | 0   |       |
| 610    | U04 |          | TEE           | 5485121   | 4                            | 10   | TEE GRV SCH STD LTCS A352 GR LCC GALV                                 | 0                           | 0   |       |
| 620    | U04 |          | TEE RED       | 5484799   | 0.75                         | 1    | TEE RED GRV X THREAD SCH 80/SCH 160 LTCS A352 GR LCC GALV             | 1.5                         | 1.5 |       |
| 630    |     |          | TEE RED       | 5484884   | 1.5                          | 3    | TEE RED GRV SCH STD/SCH 80 LTCS A352 GR LCC GALV                      | 4                           | 10  |       |
| 640    |     |          | TEE RED       | 5484881   | 4                            | 8    | TEE RED GRV SCH STD LTCS A352 GR LCC GALV                             | 6                           | 10  |       |
| 650    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 650    | U04 |          | CAP           | 5485130   | 1.5                          | 3    | CAP GRV SCH 80 LTCS A352 GR LCC GALV                                  | 0                           | 0   |       |
| 660    | U04 |          | CAP           | 5485131   | 4                            | 10   | CAP GRV SCH STD LTCS A352 GR LCC GALV                                 | 0                           | 0   |       |
| 670    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 670    | U04 |          | RED CONC      | 5484890   | 1.5                          | 2    | REDUCER CONC GRV SCH 80 LTCS A352 GR LCC GALV                         | 2                           | 3   |       |
| 680    | U04 |          | RED CONC      | 5484891   | 1.5                          | 3    | REDUCER CONC GRV SCH STD/SCH 80 LTCS A352 GR LCC GALV                 | 4                           | 6   |       |
| 690    | U04 |          | RED CONC      | 5484892   | 4                            | 8    | REDUCER CONC GRV SCH STD LTCS A352 GR LCC GALV                        | 6                           | 10  |       |
| 700    | U04 |          | RED ECC       | 5484830   | 1.5                          | 2    | REDUCER ECC GRV SCH 80 LTCS A352 GR LCC GALV                          | 2                           | 3   |       |
| 710    | U04 |          | RED ECC       | 5484831   | 1.5                          | 3    | REDUCER ECC GRV SCH STD/SCH 80 LTCS A352 GR LCC GALV                  | 4                           | 6   |       |
| 720    | U04 |          | RED ECC       | 5484832   | 4                            | 8    | REDUCER ECC GRV SCH STD LTCS A352 GR LCC GALV                         | 6                           | 10  |       |
| 730    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 730    | U04 |          |               |           |                              |      | *****COUPLINGS, UNIONS, PLUGS & THREDOLETS*****                       |                             |     |       |
| 740    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 740    | U04 |          | Coupling      | 5491250   | 0.75                         | 1    | COUPLING 3000# SCR D FNPT LTCS A350 LF2 CL.1 GALV A153                | 0                           | 0   |       |
| 750    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 750    | U04 |          | UNION         | 5491270   | 0.75                         | 1    | UNION 3000# SCR D FNPT LTCS A350 LF2 CL.1 GALV A153                   | 0                           | 0   |       |
| 760    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 760    | U04 | UD7      | FITTING SCR D | 5491280   | 0.5                          | 1    | PLUG ROUND HEAD SCR D MNPT LTCS A350 LF2 CL.1 GALV A153               | 0                           | 0   |       |
| 770    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 770    | U04 |          | FITTING SCR D | 5491325   | 0.5                          | 1.5  | BUSHING 3000# SCR D (MNPT/FNPT) HEX HEAD LTCS A350 LF2 CL.1 GALV A153 | 0.75                        | 2   |       |
| 780    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 780    | U04 | UD7      | OLETS         | 5514772   | 0.75                         | 1    | THREDOLET 3000# LTCS A350 LF2 CL.1                                    | 18                          | 24  |       |
| 790    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 790    | U04 |          |               |           |                              |      | ***** COUPLINGS, BRANCH OUTLETS - VICTAULIC *****                     |                             |     |       |
| 800    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 800    | U04 |          | Coupling      | 5484764   | 1.5                          | 16   | COUPLING RIGID GRV ASTM A352 GRADE LCC EPDM GASKET GALV               | 0                           | 0   |       |
| 810    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 810    |     |          | SADDLE        | 5485125   | 0.5                          | 2    | VICTAULIC BOLTED BRANCH OUTLET FNPT LTCS A352 GR LCC GALV             | 2                           | 16  |       |
| 820    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 820    | U04 |          |               |           |                              |      | ***** FLANGES *****                                                   |                             |     |       |
| 830    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 830    | U04 |          | FLANGE        | 5599691   | 0.5                          | 0.75 | FLG SCR D (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1 GALV A153        | 0.75                        | 2   |       |
| 840    | U04 |          | FLANGE        | 5599689   | 0.75                         | 1    | FLG SCR D (FNPT) 150# RF LTCS A350 LF2 CL.1 GALV A153                 | 0                           | 0   |       |
| 850    | U04 |          | FLANGE        | 5599701   | 0.75                         | 1    | FLG SCR D (FNPT) 300# RF LTCS A350 LF2 CL.1 GALV A153                 | 0                           | 0   |       |
| 860    | U04 |          | FLANGE        | 5556261   | 8                            | 10   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH STD                             | 0                           | 0   |       |
| 870    | U04 | FV6      | FLANGE        | 5556274   | 12                           | 12   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH 30                              | 0                           | 0   |       |
| 880    | U04 | FV6      | FLANGE        | 5556265   | 14                           | 16   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH 20                              | 0                           | 0   |       |
| 890    | U04 |          | FLANGE        | 5556265   | 18                           | 24   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH 20                              | 0                           | 0   |       |
| 900    | U04 |          | BLIND         | 5615567   | 0.5                          | 24   | FLG BLIND 150# RF LTCS A350 LF2 CL.1 GALV A153                        | 0                           | 0   |       |
| 910    | U04 |          | BLIND         | 5615577   | 0.5                          | 0.5  | FLG BLIND 300# RF LTCS A350 LF2 CL.1 GALV A153                        | 0                           | 0   |       |
| 920    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 920    | U04 |          |               |           |                              |      | ***** FLANGES - VICTAULIC *****                                       |                             |     |       |
| 930    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 930    | U04 | FS1      | FLANGE        | 5754404   | 1.5                          | 16   | FLANGE ADAPTER NIPPLE 150# RF / GRV A352 GR LCC GALV                  | 0                           | 0   |       |
| 940    | U04 | FS1      | FLANGE        | 5754405   | 1.5                          | 16   | FLANGE ADAPTER NIPPLE 300# RF / GRV A352 GR LCC GALV                  | 0                           | 0   |       |
| 950    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 950    | U04 |          |               |           |                              |      | ***** ORIFICE FLANGES *****                                           |                             |     |       |
| 960    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 960    | U04 |          | ORIFICE FLG   | 5623941   | 18                           | 24   | WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 20                         | 0                           | 0   |       |
| 970    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 970    | U04 |          |               |           |                              |      | ***** ORIFICE FLANGE - VICTAULIC *****                                |                             |     |       |
| 980    | U04 |          |               |           |                              |      | ***** PIPE NIPPLES *****                                              |                             |     |       |
| 990    |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 990    | U04 |          | NIPPLE        | 5541200   | 0.5                          | 1    | NIPPLE SCH 160 LTCS A333 GR.6 TBE 3" LG (MNPT) GALV A153              | 0                           | 0   |       |
| 1000   | U04 |          | NIPPLE        | 5541201   | 0.5                          | 1    | NIPPLE SCH 160 LTCS A333 GR.6 TBE 7" LG (MNPT) GALV A153              | 0                           | 0   |       |
| 1010   |     |          |               |           |                              |      |                                                                       |                             |     |       |
| 1010   | U04 |          |               |           |                              |      | ***** SWAGE NIPPLES *****                                             |                             |     |       |
| 1020   |     |          |               |           |                              |      |                                                                       |                             |     |       |

|  | <b>TENGIZCHEVROI<br/>TCO<br/>TENGIZ,<br/>KAZAKHSTAN</b> | CONSTRUCTION                                                                                                                                                        | CORR.ALLOW.       | SERVICE                                                                 | MATERIAL                                                                                       | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|                                                                                    |                                                         | 3/4"-1"- SCREWED<br>1 1/2" - 16" - VICTAULIC GROOVE & COUPLING<br>18" & ABOVE BUTTWELDED<br>SEAL WELDING - NONE<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW. 3 MM | DESIGN: 15.8 BARG<br>CODE - ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS VICTAULIC COUPLINGS<br>FLANGE RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 110 °C | 12/16/2024 | BY<br>OZAC | U04 | Page 2<br>of 3 | <b>150VC01</b>    |

| SEQ NO | REV | NOTE NO. | ITEM NAME  | ITEM CODE | SIZE RANGE / SMALL END SIZES |    | DESCRIPTION                                            | REFERENCE / LARGE END SIZES |     | NOTES |
|--------|-----|----------|------------|-----------|------------------------------|----|--------------------------------------------------------|-----------------------------|-----|-------|
| 1020   | U04 |          | RED CONC   | 5544150   | 0.75                         | 1  | SWAGE CONC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153 | 1                           | 1.5 |       |
| 1030   | U04 |          | RED ECC    | 5549150   | 0.75                         | 1  | SWAGE ECC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153  | 1                           | 1.5 |       |
| 1040   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1040   | U04 |          |            |           |                              |    | ***** GASKETS *****                                    |                             |     |       |
| 1050   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1050   | U04 |          | GASKET     | 5673043   | 0.5                          | 24 | GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR             | 0                           | 0   |       |
| 1060   | U04 | FS1      | GASKET     | 5673033   | 0.5                          | 12 | GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR             | 0                           | 0   |       |
| 1070   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1070   | U04 |          |            |           |                              |    | ***** GASKETS - VICTAULIC *****                        |                             |     |       |
| 1080   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1080   | U04 |          | GASKET     | NOMTO_G   | 1.5                          | 16 | GASKET FOR CLAMP FOR VICTAULIC SYSTEM - NON REPORTABLE | 0                           | 0   |       |
| 1090   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1090   | U04 | B1       |            |           |                              |    | ***** BOLTING *****                                    |                             |     |       |
| 1100   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1100   | U04 |          | BOLTING    | 5675770   | 0.5                          | 99 | STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS      | 0                           | 0   |       |
| 1110   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1110   | U04 |          |            |           |                              |    | **** BLINDS & SPACERS ****                             |                             |     |       |
| 1120   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1120   | U04 |          | SPEC BLIND | 5749582   | 0.75                         | 14 | SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153     | 0                           | 0   |       |
| 1130   | U04 |          | SPEC BLIND | 5749583   | 0.75                         | 10 | SPECTACLE BLIND 300# RF LTCS A516 GR.70 GALV A153      | 0                           | 0   |       |
| 1140   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1140   | U04 |          | PADDLE     | 5749942   | 16                           | 16 | PADDLE BLANK 150# RF LTCS A516 GR.70 GALV A153         | 0                           | 0   |       |
| 1150   | U04 |          | PADDLE     | 5750195   | 16                           | 16 | PADDLE SPACER 150# RF LTCS A516 GR.70 GALV A153        | 0                           | 0   |       |
| 1160   | U04 |          | PADDLE     | 5750023   | 18                           | 24 | PADDLE BLANK 150# RF LTCS A516 GR.70                   | 0                           | 0   |       |
| 1170   | U04 |          | PADDLE     | 5750240   | 18                           | 24 | PADDLE SPACER 150# RF LTCS A516 GR.70                  | 0                           | 0   |       |
| 1180   | U04 |          | PADDLE     | 5749853   | 12                           | 24 | PADDLE BLANK 300# RF LTCS A516 GR.70 GALV A153         | 0                           | 0   |       |
| 1190   | U04 |          | PADDLE     | 5750201   | 12                           | 24 | PADDLE SPACER 300# RF LTCS A516-GR70 GALV A153         | 0                           | 0   |       |
| 1200   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1200   | U04 |          |            |           |                              |    | ***** MISCELLANEOUS*****                               |                             |     |       |
| 1210   |     |          |            |           |                              |    |                                                        |                             |     |       |
| 1210   | U04 |          | GASKET     | 5707599   |                              |    | LUBRICANT FOR GASKET PUSH-ON CI & DI                   |                             |     |       |
| 1220   |     |          |            |           |                              |    |                                                        |                             |     |       |

|                                                                                    |                                                |                                                                                                                                                                       |                   |                                                                         |                                                                                                |            |            |     |                |                   |
|------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|  | TENGIZCHEVROIL<br>TCO<br>TENGIZ,<br>KAZAKHSTAN | CONSTRUCTION                                                                                                                                                          | CORR.ALLOW.       | SERVICE                                                                 | MATERIAL                                                                                       | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|                                                                                    |                                                | 3/4"- 1" - SCREWED<br>1 1/2" - 16" - VICTAULIC GROOVE & COUPLING<br>18" & ABOVE BUTTWELDED<br>SEAL WELDING - NONE<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW. 3 MM | DESIGN: 15.8 BARG<br>CODE - ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS VICTAULIC COUPLINGS<br>FLANGE RATING: 150 RF<br>TEMPERATURE RANGE:- 40 / 110 °C | 12/16/2024 | BY<br>OZAC | U04 | Page 3<br>of 3 | 150VC01           |